

250908 M | Functions

heretofor

We have used functions that other coders have written for us, but we haven't yet written our own. We also learned about `if` statements, and `while` and `for` loops, but we need to practice with these concepts some more

herein

Now that we have talked through the major operators in python, we need to put this altogether with the python function. Functions in python are small packages of code that can take in input and execute a series of steps based on that input, and optionally return an output. Functions are defined beginning with `def` statements so something like this:

```
def myFunction(inputs, go, here, if, necessary):  
    print('this is a function')  
    output = inputs*go*here**(if+necessary)  
    return(output)
```

Now, we can write functions that can perform a particular calculation with different inputs, and we can begin to keep some of these so we can refer to them again later.

We wrote a couple of functions in class, one for calculating factorials and another for understanding spherical coordinates. For homework do exercise 2.11 and take an honest crack at 2.12.

hence

Now that we have functions down, we are ready to move on to plotting. After we learn how to produce some simple plots, we will be ready to tackle most of the rest of the subject of computational physics.

```
In [1]: def myFunction():  
        print('this is a function')
```

```
In [3]: myFunction()
```

this is a function

1. make a factorial function

2. make it return the factorial for any number (use a for loop!)
3. don't test it too high!

```
In [36]: def factorial(x):  
        f = 1  
        if x<0:  
            print('undefined')  
        else:  
            for i in range(1,x+1):  
                f = f*i  
  
        return(f)
```

```
In [37]: y = factorial(5)
```

```
In [35]: y
```

```
Out[35]: 120
```

```
In [43]: def distance(x,y,z):return((x**2+y**2+z**2)**(1/2))
```

```
In [44]: distance(3,0,0)
```

```
Out[44]: 3.0
```

```
In [45]: def sph(r, theta, phi):  
        from math import cos, sin  
        x = r*sin(theta)*cos(phi)  
        y = r*sin(theta)*sin(phi)  
        z = r*cos(theta)  
        return(x,y,z)
```

```
In [46]: from math import pi
```

```
sph(1,pi/2, 0)
```

```
Out[46]: (1.0, 0.0, 6.123233995736766e-17)
```

```
In [ ]:
```